

Internal Assessment - 02

SECTION A – THEORY

Answer all questions. Each question carries 2 marks.

1. Compare HDD and SSD from a forensics perspective. Why is data recovery more difficult on modern SSDs?
2. Explain the difference between slack space, unallocated space, and free space. How does each become a source of forensic evidence?
3. What is a write-blocker? Explain hardware write-blockers vs software write-blockers. When would you use each?

SECTION B – PRACTICAL

Answer the question below. It carries 4 marks.

1. On a Windows system, perform the following steps:
 - a) Open 5 different files — at least one Word document (.docx), one PDF, and one image file (JPG/PNG). Take a screenshot showing all 5 files open. **(1 Mark)**
 - b) Use **FTK Imager** to capture a live RAM dump of the running system. Write the steps followed during acquisition. **(1 Mark)**
 - c) Load the RAM dump in **Volatility**. Identify the correct OS profile and list all running processes at the time of capture. Write the exact commands used. **(1 Mark)**
 - d) Run the appropriate Volatility plugin to find evidence of the opened files in memory. List the file names identified in the output and explain how this serves as forensic evidence. **(1 Mark)**

— END OF QUESTION PAPER —